



5. Covariance Matrix and Correlation Matrix

Two of the most powerful matrices in Machine Learning, and significantly Data Science

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Covariance Matrix and Correlation Matrix are two of the most powerful matrices in Machine Learning, and honestly, two of the most misunderstood.

They appear everywhere. In statistics, in data science, in algorithms you haven't even realized are using them. But here's the strange part: Most people don't really *see* them. They memorize the formulas. They plug them into libraries. They move on. And just like that, something incredibly important becomes background noise.

But you're not here for that. You're here to understand what these matrices are actually *doing*. So let's slow this down.

Think about what Machine Learning really tries to do. It looks at data, and asks a very quiet question: *"How are things related?"*

Not just individually. But together. Because a single feature rarely tells the whole story.

- Height alone doesn't define a person.
- Price alone doesn't define a house.
- One pixel doesn't define an image.

Meaning only appears when things start interacting. And that's where these matrices come in. They don't look at features in isolation. They observe, they measure, they document relationships. Not in words. Not in rules. But in structure.

If I were the model itself, I wouldn't ask: **"What is this feature?"** → I would ask: **"How does this feature move with others?"** → **"When one changes, what happens to the rest?"**

Because that's where the truth is. Not in the values themselves, but in their behavior.

So in this section, we're going to explore two perspectives of the same idea:

- One that captures **how features vary together**
- And one that refines that idea into something more **interpretable**

We start with the raw observation. No normalization. No scaling. Just pure measurement.

1. Covariance Matrix: Mapping the Magnitude of Relationships

A Covariance Matrix sounds technical. And yes, it is. But at its core? It's just a way of **observing how variables move together**.

Imagine myself standing in a crowded scene. I don't just look at one person. I watch *everyone*. Not individually, but in relation.

- When one moves forward, who follows? And when one slows down, who drifts away?

Because movement alone is meaningless. **Synchronized movement?** That's information.

That's exactly what a Covariance Matrix does. It doesn't ask: "What is this variable?" → It asks: *"How does this variable change, when another one changes?"*

Formally, a Covariance Matrix is a **square matrix** that contains the covariances between a set of variables. But don't let that definition trap you.

Think of it as a structured map of relationships → A complete snapshot of how every variable interacts with every other. Not one relationship. Not a few. **All of them. At once.**

Symmetry — The Quiet Balance

Here's something interesting. The Covariance Matrix is always **symmetric**. Meaning: $\text{Cov}(X, Y) = \text{Cov}(Y, X)$

But I wouldn't call this a formula. I would call it obvious. If X moves with Y , then Y must move with X . Relationships don't have direction. They have **mutual behavior**. So the matrix reflects that.

Everything above the diagonal is mirrored below it. Which means you don't need to observe everything twice. Half the work already tells the full story

Diagonal vs Off-Diagonal — Two Perspectives

Now look closer. Not all elements in the matrix mean the same thing. **The diagonal elements?** They're different. They don't describe relationships between variables. They describe something more internal: How much a variable varies *by itself*.

In other words: $\text{Cov}(X, X) = \text{Var}(X)$

I would interpret it like this: *"How unstable is this variable... on its own?"*

Then there are the **off-diagonal elements**. This is where things get interesting. These values answer: *"Do these two variables move together... or against each other?"*

- Positive covariance → They rise and fall together
- Negative covariance → One rises while the other falls
- Near zero → They don't really care about each other

This is not just data. This is **behavior**.

Size — When Observation Scales

If there are n variables... The Covariance Matrix becomes: $n \times n$. Which means: *Every variable is compared with every other variable*

The number of elements grows quickly. n^2 total entries

But because of symmetry? You only get: $\frac{n(n+1)}{2}$ unique relationships. Even then, as the number of variables increases, the complexity doesn't just grow. It **explodes**.

A Simple Case — Two Variables

Let's shrink the world. Just two variables: X and Y . Then the Covariance Matrix becomes:

$$\begin{bmatrix} \text{Cov}(X, X) & \text{Cov}(X, Y) \\ \text{Cov}(Y, X) & \text{Cov}(Y, Y) \end{bmatrix}$$

I would read this like a report:

- Top-left → How X behaves alone
- Bottom-right → How Y behaves alone
- The other two → How X and Y interact

The Mechanics — What It Actually Measures.

Underneath all of this, covariance is measuring something very specific: *How deviations from the mean align*

For a variable X and a variable Y :

$$\text{Cov}(X, X) = \text{Var}(X) = \frac{\sum (X_i - \bar{X})^2}{n - 1} ; \text{Cov}(Y, Y) = \text{Var}(Y) = \frac{\sum (Y_i - \bar{Y})^2}{n - 1}$$

For two variables X and Y :

$$\text{Cov}(X, Y) = \frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{n - 1}$$

I wouldn't focus on the formula. But would focus on the idea: "When X deviates from its normal... does Y follow?". Because that's what this really is. Not numbers. Not equations. But **coordinated deviation**.

And once you see it like that, the covariance matrix stops being a block of numbers. It becomes something else entirely. A record of how variables **behave together**.

Example: Observing How Two Subjects Move Together

Let's make this real. Not theory. Not abstraction. Just three students.

Student	Math (X)	Science (Y)
1	92	80
2	60	30
3	100	70

We don't start with relationships. We start with **baselines**. Because before asking *how things move together*... we need to know where they normally stand.

- **Step 1 — Find the Means**

$$\bar{X} = \frac{92 + 60 + 100}{3} = 84 ; \bar{Y} = \frac{80 + 30 + 70}{3} = 60$$

These are not just averages. They are reference points. Where each subject “rests”. The center of their behavior.

- **Step 2 — How Each Variable Moves Alone**

Now we observe variability. Not relationships yet. Just... instability.

$$\sum (X_i - \bar{X})^2 = (92 - 84)^2 + (60 - 84)^2 + (100 - 84)^2 = 896$$

$$\Rightarrow \text{Cov}(X, X) = \frac{896}{3 - 1} = 448$$

$$\sum (Y_i - \bar{Y})^2 = (80 - 60)^2 + (30 - 60)^2 + (70 - 60)^2 = 1400$$

$$\Rightarrow \text{Cov}(Y, Y) = \frac{1400}{3 - 1} = 700$$

What does this tell us? Science varies more than Math → Its spread is wider... less stable

- **Step 3 — Deviations (The Real Signals)**

Now we look at how each value *deviates* from its mean:

$$\text{Student 1: } X - \bar{X} = 8, \quad Y - \bar{Y} = 20$$

$$\text{Student 2: } X - \bar{X} = -24, \quad Y - \bar{Y} = -30$$

$$\text{Student 3: } X - \bar{X} = 16, \quad Y - \bar{Y} = 10$$

These numbers are not random. They are direction: **Positive** → **above average**; **Negative** → **below average**

- **Step 4 — How They Move Together**

Now comes the real question: *When Math shifts... what does Science do?*

$$\sum (X_i - \bar{X})(Y_i - \bar{Y}) = (8 \times 20) + (-24 \times -30) + (16 \times 10) = 1040$$

And because relationships are mutual:

$$\text{Cov}(X, Y) = \text{Cov}(Y, X) = \frac{1040}{2} = 520$$

- **Step 5 — Construct the Covariance Matrix**

Now we stop calculating... as there is no longer anything to calculate. And start **seeing the structure**.

$$\begin{bmatrix} 448 & 520 \\ 520 & 700 \end{bmatrix}$$

This matrix is small. But it says a lot: 448 → Math varies but not wildly; 700 → Science varies more; 520 → They move **together**, strongly. Not perfectly. But clearly.

And that's the key insight. We didn't analyze Math alone. We didn't analyze Science alone. We observed how they **coexist**.

2. Correlation Matrix: Standardized Strength at a Glance

The covariance matrix told us something important: *How variables move together*. But it also had a problem. It speaks in **raw magnitude**. And raw magnitude can be misleading.

Imagine two relationships: **One measured in centimeters** → **One measured in kilometers**. Even if they behave the same way... their covariance values will look completely different. Not because the relationship changed. But because the **scale changed**.

So we ask a better question:

┃ *What if we remove the scale... and keep only the relationship?*

That's where the correlation matrix comes in.

It is not a replacement. It is a **refinement**. A normalized version of covariance. One that doesn't care about units. Only about **strength and direction**.

If covariance is observation, then correlation is **interpretation**.

What It Actually Does - The correlation matrix takes every covariance and scales it. So that all relationships fall into one simple range: -1 to 1 → No more ambiguity. No more unit confusion. Just clarity.

Structure — Familiar, But Cleaner

Just like covariance, the correlation matrix is **symmetric**: $\text{Corr}(X, Y) = \text{Corr}(Y, X)$ → Because relationships don't change direction.

But there's one key difference. The diagonal is no longer "variance". It becomes something absolute: $\text{Corr}(X, X) = 1$

Always. Because every variable is perfectly correlated with itself.

So the correlation matrix now looks like this:

- Diagonal → always 1
- Off-diagonal → relationships between variables

Clean. Consistent. Comparable.

Interpreting the Values

Now the numbers finally *mean something intuitive*.

- Close to 1 → Strong positive relationship
- Close to -1 → Strong negative relationship
- Around 0 → No meaningful linear relationship

But we can go deeper. Think of it like reading behavior:

Positive Correlation

- 0.00 : No relationship
- 0.01 → 0.19 : Very weak
- 0.20 → 0.39 : Weak
- 0.40 → 0.59 : Moderate
- 0.60 → 0.79 : Strong
- 0.80 → 0.99 : Very strong
- 1.00 : Perfect alignment

Negative Correlation mirrors this in the opposite direction.

- 1.00 : Perfect negative alignment

- $-0.99 \rightarrow -0.80$: Very strong negatively
- $-0.79 \rightarrow -0.60$: Strong negatively
- $-0.59 \rightarrow -0.40$: Moderate negatively
- $-0.39 \rightarrow -0.20$: Weak negatively
- $-0.19 \rightarrow -0.01$: Very weak negatively

This is why correlation feels more natural. Because it doesn't just give numbers. It gives **meaning**.

The Formula — What Makes It Work: Underneath, correlation is just covariance... adjusted.

$$\text{Corr}(X, Y) = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

- Where: σ_X, σ_Y are standard deviations $\rightarrow$ The square roots of variance

What this does is simple—but powerful: It divides out the scale. It keeps only the relationship

So no matter the units... The result always stays between -1 and 1 .

Why This Matters - Covariance tells you: *"These variables move together... by this amount"* $\rightarrow$ Correlation tells you: *"These variables are related... by this strength"*

One is raw observation. The other is **comparable insight**.

And that's why, in practice, when you want to *understand relationships clearly*, you don't just observe. You normalize.

Example: Calculate the Covariance Matrix and Correlation Matrix of the given data.

Let's step into a slightly bigger world. Not just two variables... but three: **Height (X); Weight (Y); Age (Z)**

Height	Weight	Age
5.50	150	25
6.00	180	30
5.70	160	28
5.90	170	35

At first glance, this is just data. But that's never the point. The real question is: *Do these variables move together? And if they do... how strongly?*

- **Step 1 — Establish the Baseline**

Before anything else... we find where each variable "rests":

$$\bar{X} = 5.775, \quad \bar{Y} = 165, \quad \bar{Z} = 29.5$$

And now... everything becomes relative.

- **Step 2 — Observe Deviations**

We don't care about raw values anymore. We care about **how far each point is from its mean**:

$X_i - \bar{X}$	$Y_i - \bar{Y}$	$Z_i - \bar{Z}$
-0.275	-15	-4.5
0.225	15	0.5
-0.075	-5	-1.5
0.125	5	5.5

These are not just differences. They are **directional signals**.

- **Step 3 — Variability (Diagonal Elements)**

Now we measure how each variable behaves on its own.

$$\sum (X_i - \bar{X})^2 = (-0.275)^2 + (0.225)^2 + (-0.075)^2 + (0.125)^2 = 0.1475$$

$$\Rightarrow \text{Cov}(X, X) = \frac{0.1475}{3} = 0.04916$$

$$\sum (Y_i - \bar{Y})^2 = (-15)^2 + (15)^2 + (-5)^2 + (5)^2 = 500$$

$$\Rightarrow \text{Cov}(Y, Y) = \frac{500}{3} = 166.67$$

$$\sum (Z_i - \bar{Z})^2 = (-4.5)^2 + (0.5)^2 + (-1.5)^2 + (5.5)^2 = 53$$

$$\Rightarrow \text{Cov}(Z, Z) = \frac{53}{3} = 17.67$$

Immediately, something becomes clear: Weight varies the most → Height varies the least. Different scales. Different spreads.

- **Step 4 — Relationships (Off-Diagonal Elements)**

Now the real observation begins. And because relationships are mutual:

$$\sum (X_i - \bar{X})(Y_i - \bar{Y}) = (-0.275 * -15) + (0.225 * 15) + (-0.075 * -5) + (0.125 * 5) = 8.5$$

$$\Rightarrow \text{Cov}(X, Y) = \text{Cov}(Y, X) = \frac{8.5}{3} = 2.833$$

$$\sum (X_i - \bar{X})(Z_i - \bar{Z}) = (-0.275 * -4.5) + (0.225 * 0.5) + (-0.075 * -1.5) + (0.125 * 5.5) = 2.15$$

$$\Rightarrow \text{Cov}(X, Z) = \text{Cov}(Z, X) = \frac{2.15}{3} = 0.7167$$

$$\sum (Y_i - \bar{Y})(Z_i - \bar{Z}) = (-15 * -4.5) + (15 * 0.5) + (-5 * -1.5) + (5 * 5.5) = 110$$

$$\Rightarrow \text{Cov}(Y, Z) = \text{Cov}(Z, Y) = \frac{110}{3} = 36.67$$

- **Step 5 — The Covariance Matrix**

Now everything comes together:

$$\begin{bmatrix} 0.04916 & 2.833 & 0.7167 \\ 2.833 & 166.67 & 36.67 \\ 0.7167 & 36.67 & 17.67 \end{bmatrix}$$

This matrix tells a story. But it's still raw. The scales are different. The magnitudes are uneven. So comparing relationships directly? Not so easy.

- **Step 6 — Normalize the View (Standard Deviations)**

We extract the scale of each variable:

$$\sigma_X = \sqrt{0.04916} = 0.2215, \quad \sigma_Y = \sqrt{166.67} = 12.91, \quad \sigma_Z = \sqrt{17.67} = 4.20$$

These are the “weights” of each variable’s variation.

- **Step 7 — Correlation (Pure Relationships)**

Now we remove the scale. We keep only the **relationship strength**.

$$\text{Corr}(X, X) = \text{Corr}(Y, Y) = \text{Corr}(Z, Z) = 1$$

$$\text{Corr}(X, Y) = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} = \frac{2.833}{0.2215 \times 12.91} = 0.98$$

$$\text{Corr}(X, Z) = \frac{\text{Cov}(X, Z)}{\sigma_X \sigma_Z} = \frac{0.7167}{0.2215 \times 4.20} = 0.77$$

$$\text{Corr}(Y, Z) = \frac{\text{Cov}(Y, Z)}{\sigma_Y \sigma_Z} = \frac{36.67}{12.91 \times 4.20} = 0.68$$

- **Step 8 — The Correlation Matrix**

$$\begin{bmatrix} 1 & 0.99 & 0.77 \\ 0.99 & 1 & 0.68 \\ 0.77 & 0.68 & 1 \end{bmatrix}$$

Now the picture is clear.

- Height and Weight: 0.99 → **almost perfectly aligned**
- Height and Age: 0.77 → **strong relationship**
- Weight and Age: 0.68 → **moderate to strong**

No units. No confusion. Just relationships.

And that’s the transformation. From raw numbers to structured behavior, and to something you can actually **understand at a glance**.

3. Covariance Matrix and Correlation Matrix with Python

- Code 1: Covariance Matrix and Correlation Matrix using the Pandas library


```
# Covariance Matrix and Correlation Matrix with Python
import pandas as pd
sample_data = {'Height': [5.50, 6.00, 5.70, 5.90],
               'Weight': [150, 180, 160, 170],
               'Age': [25, 30, 28, 35]}
# Convert to data frame
df = pd.DataFrame(sample_data)
cov_matrix = df.cov()
corr_matrix = df.corr()
print("Covariance Matrix:\n", cov_matrix)
print("Correlation Matrix:\n", corr_matrix)
```

Covariance Matrix:

	Height	Weight	Age
Height	0.049167	2.833333	0.716667
Weight	2.833333	166.666667	36.666667
Age	0.716667	36.666667	17.666667

Correlation Matrix:

	Height	Weight	Age
Height	1.000000	0.989778	0.768961
Weight	0.989778	1.000000	0.675725
Age	0.768961	0.675725	1.000000

- Code 2: Covariance and Correlation Heatmaps

```
# Covariance Matrix and Correlation Matrix with Python
# Heatmaps
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

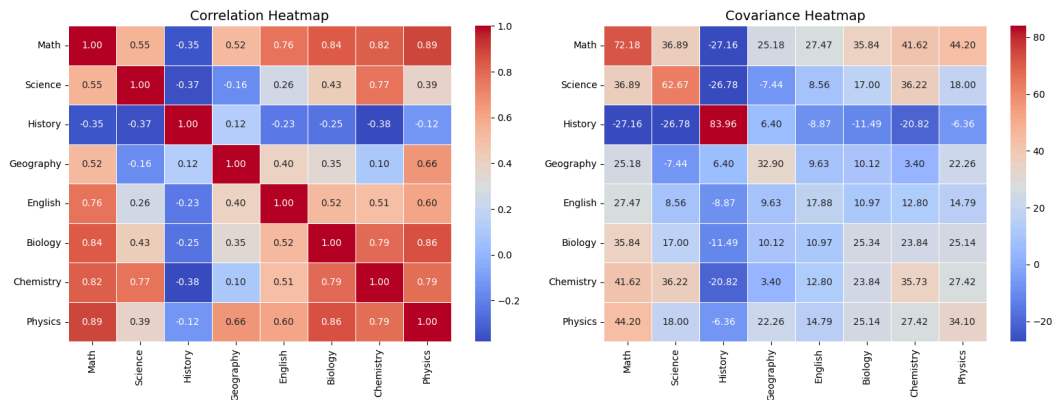
sample_data = {'Math': [84, 82, 81, 89, 73, 94, 92, 70, 88, 95],
               'Science': [85, 82, 72, 77, 75, 89, 95, 84, 77, 94],
               'History': [97, 94, 93, 95, 88, 82, 78, 84, 69, 78],
               'Geography': [88, 85, 90, 92, 80, 79, 84, 75, 89, 91],
               'English': [90, 88, 86, 85, 84, 93, 87, 78, 91, 89],
               'Biology': [82, 79, 85, 88, 81, 90, 92, 76, 83, 87],
               'Chemistry': [79, 85, 80, 83, 78, 91, 94, 77, 81, 90],
               'Physics': [84, 86, 88, 90, 79, 88, 93, 74, 85, 92]}

# Convert to data frame
df = pd.DataFrame(sample_data)
cov_matrix = df.cov()
corr_matrix = df.corr()

# Set up the matplotlib figure with two subplots
fig, axes = plt.subplots(1, 2, figsize=(16, 6))
# Draw the correlation heatmap
sns.heatmap(corr_matrix, annot=True, cmap='coolwarm', fmt=".2f",
            linewidths=0.5, ax=axes[0])
axes[0].set_title("Correlation Heatmap", fontsize=14)
# Draw the covariance heatmap
```

```
sns.heatmap(cov_matrix, annot=True, cmap='coolwarm', fmt=".2f",
             linewidths=0.5, ax=axes[1])
axes[1].set_title("Covariance Heatmap", fontsize=14)

# Adjust layout and show plot
plt.tight_layout()
plt.show()
```



Summary

Covariance Matrix and Correlation Matrix are simple. Almost *too* simple. They don't try to predict. They don't try to optimize. They don't try to be clever → They just **observe relationships**.

But here's the part people underestimate: Simplicity doesn't mean unimportant. It means **foundational**.

Because behind so many "advanced" techniques:

- PCA → built on covariance
- LDA → depends on class covariance
- Gaussian models → defined by covariance
- Feature selection → guided by correlation

They all start here. If you don't understand this, everything else becomes mechanical. You compute, but you don't *see*.

Covariance and Correlation are simple. But they are the kind of simple that quietly supports everything else. The kind you don't notice, until you realize **everything depends on it**.

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Next Section:

 [6. Overfitting and Underfitting](#)